CAIE IGCSE Chemistry

12.2 Acid–base titrations

Notes

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Describe an acid–base titration to include the use of a: (a) burette (b) volumetric pipette (c) suitable indicator

Titrations are carried out to calculate the concentration of an acid or an alkali by determining how much acid is needed to neutralise an alkali (or vice versa). Titrations can also be done to prepare a soluble salt from the reaction between an acid and dilute solution of alkali

Apparatus required

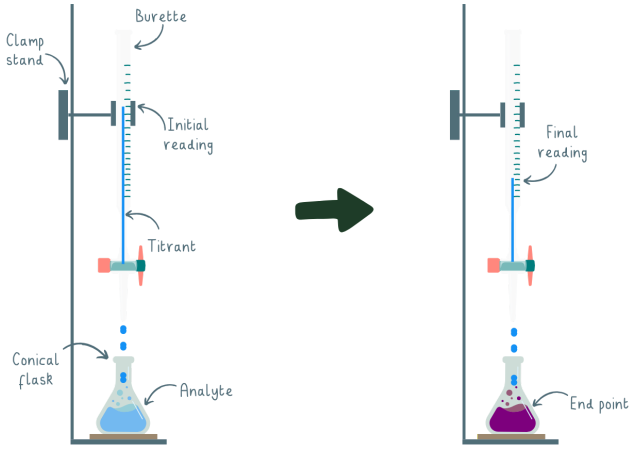
- 50cm^3 burette
- 25cm^3 volumetric pipette
- Pipette filler
- 250cm^3 conical flask
- Suitable indicator, e.g. phenolphthalein
- Sodium hydroxide solution of concentration 0.1mol/dm^3
- Sulfuric acid of unknown concentration
- Clamp stand
- White tile
- Funnel

Method

1. Use the pipette filler to draw up 25cm^3 of sodium hydroxide solution into the volumetric pipette and add this to the conical flask.
2. With the burette clamped onto the stand, use the funnel to fill the burette with sulfuric acid.
3. Place an empty beaker underneath the tap and release a small amount of the acid to prevent any air bubbles.
4. Record the initial volume on the burette to the nearest 0.05cm^3, taking care to ensure that you are eye-level to the meniscus.
5. Place the conical flask underneath the tap on top of a white tile to allow any colour changes to be more visible.
6. Add 3 drops of the phenolphthalein indicator into the conical flask and swirl gently to mix (phenolphthalein is pink when alkaline and colourless when acidic).
7. Run the acid from the burette into the alkali, whilst swirling the solution. As the endpoint is approached (the colour of the solution becomes pale pink), reduce the increments and quickly stop the titration when it is reached.
8. Record the final volume on the burette in your table to the nearest 0.05cm^3.
9. Calculate the rough titre using the following equation:
   Final volume - Initial volume = Titre volume
10. Repeat the experiment until 2 concordant titres are found (results that are within 0.1cm^3 of each other excluding the rough titration).

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11. Calculate the mean titre volume using the 2 concordant titre volumes.



Example table of results

Titration	Initial volume /cm^3	Final volume /cm^3	Titre /cm^3
Rough	0.20	24.65	24.45
1	0.35	25.20	24.85
2	1.10	25.70	24.60
3	0.90	25.85	24.95

Mean titre: (24.85 + 24.95) / 2 = 24.9 cm^3

Describe how to identify the end-point of a titration using an indicator

Indicators are used to identify the end-point of a titration as a sudden colour change will occur when the solution goes from acid to alkali (or vice versa). The two most commonly used acid-base indicators are:

1. Phenolphthalein: pink in alkali -> colourless in acid
2. Methyl orange: yellow in alkali -> red in acid

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